

Virtualization Game

Level 1: Step 2
EASY
Theme: Virtualization and Operating Systems
Due Date: 08/10/2022

1. **Instructional Objective:** Virtualization Game is a card game to help students identify and learn the relational structure of knowledge related to virtualization and operating systems. It allows building concept connection skills and critical thinking, as well as helping the student to research and systematize.

2. The suggested ruleset for the game is as follows:

(Students are invited to tweak the rules as desired...)

- The deck consists of $52 + 1$ cards. There are three types of cards:
 - 1 card containing a short set of instructions on how to play;
 - **VM Card**: this type represents the goal of the game - to collect virtual machines;
 - **Incident Card**: describes what the player should do. Can be a question, a challenge, a trap...
- After removing the card with the game instructions, the deck is shuffled.
- The cards are scattered in the middle of the table.
- After deciding who will start, that person will draw a card from the pile:
 - If that card is a **VM Card**, it's placed in front of him, and he can continue or skip turn;
 - If the card is an **Incident Card**, all other the players should make sure the player answers the question or faces the consequence (If wrong...).

Afterwards, the card is placed in the pile and mixed with the others.

- The gameplay continues until a player collects 10 VM Cards.
- The point of equilibrium is somewhere between 20 and 25 VM cards, although it depends on the chosen game mechanics

↑ VMs → game ends quicker / ↓ VMs → game takes longer

3. **Card Design (Figure 1):**

Back Design - Does not contain any information that reveals the Incident/VM.

Front Design - Represents a VM or contains an Incident in the form of a picture, chart, question, instruction, ...

4. Deliverables (all deliverables submitted via [discord#assignments-submission](#)):

1. A .zip file containing the files¹:
 - vm.png
 - front.png
 - back.png
 - incidents.xlsx (Table 1)
2. The file name should be 11s2e_aXXXXX_aYYYYY_aZZZZZ_aVVVVV.zip and it should contain all the files of the deliverables. This code means:
 - Level 1
 - Step 2
 - Easy difficulty
 - Logins of the Students that participated in the group separated by ' _ '
 - **Maximum:** 4 students

Table 1: Example of the incidents.xlsx file structure.

Title	Text	If wrong. . .
Hypervisor	What is the main difference between a type 1 hypervisor and a type 2?	Return one VM.
VMM	What is the meaning and responsibility of a VMM?	You're out for 1 round.
...	...	...

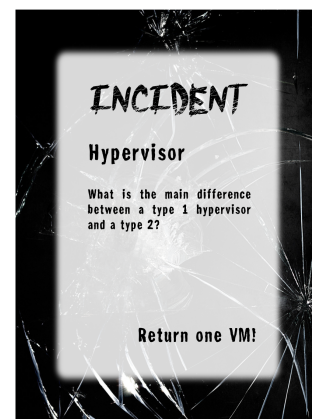
The solution must not be included in the excel file. . .



(a) Back of all cards.



(b) VM Card.



(c) Incident Card.

Figure 1: Sample card template design.

This level is mandatory for all students!

¹ .rar files will be discarded!